

CEO Dismissals and Stock Market Reactions

An Event Study Approach with Machine Learning

DSA 210 — Introduction to Data Science
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Final Report

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1. Introduction & Motivation

When a CEO is forced out of a company, it sends a direct signal to the market. Some dismissals are welcomed by investors as a necessary course correction; others create uncertainty and drive the stock price down. Understanding this dynamic has implications both for corporate governance research and for evaluating whether a short-term trading opportunity exists around these events.

This project investigates how stock prices of S&P; 1500 companies react to forced CEO departures using an **event study methodology**. By calculating Cumulative Abnormal Returns (CAR) around dismissal dates and testing whether the reaction varies by departure type, crisis periods, and historical trends, I aim to quantify the market's response to CEO turnover. Additionally, machine learning models are applied to assess whether the direction and magnitude of the market reaction can be predicted from observable features.

Research Questions:

1. Does the stock market exhibit a statistically significant reaction to forced CEO departures?
2. Does the reaction vary by departure type, crisis period, or historical era?
3. Can machine learning models predict the direction or magnitude of the reaction?

2. Data Source & Characteristics

The primary dataset is the **CEO Dismissal Database** compiled by Gentry, Harrison, Quigley & Boivie (2021), published in the *Strategic Management Journal*, Vol. 42(5), pp. 968-991. The dataset covers 9,390 CEO departure records across 3,860 unique S&P; 1500 companies from 1992-2019, with a binary `ceo_dismissal` flag indicating whether the departure was forced.

To study stock market reactions, the dataset was enriched by: (1) mapping company identifiers (`gvkey`) to stock tickers via manual matching and notes extraction, achieving a 29.6% match rate for dismissal events; (2) downloading daily stock prices around each event using `yfinance`; and (3) calculating abnormal returns using the market model with S&P; 500 as benchmark.

Metric	Value
Total departures in dataset	9,390
Forced dismissals (<code>ceo_dismissal=1</code>)	1,414
Dismissals with matched tickers	405
Valid CAR calculations	204
Year range	1995 - 2019
Unique companies in sample	~160

Events with $|CAR| > 100\%$ were excluded as data errors (1 event), leaving 204 clean observations for hypothesis testing and ML analysis.

3. Methodology

3.1 Event Study Design

For each CEO dismissal event, a classic event study was conducted. The **estimation window** (Day -150 to Day -21) was used to estimate expected returns via ordinary least squares regression: $R_{\text{stock}} = \alpha + \beta \times R_{\text{market}}$. The **event window** (Day -5 to Day +20) was used to calculate abnormal returns (AR = actual return minus expected return) and cumulative abnormal returns (CAR) over multiple windows:

CAR[0,+2] for immediate reaction, CAR[0,+5] for one-week, CAR[3,+20] for post-reaction drift, and CAR[-5,+20] for the full window.

3.2 Statistical Tests

Hypothesis	Test	Null Hypothesis
H1: Negative CAR	One-sample t-test	Mean CAR[0,+2] = 0
H2: Code 3 vs 4	Mann-Whitney U	No distribution difference
H3: Mean reversion	Spearman correlation	$\rho = 0$
H4: Crisis effect	Mann-Whitney U (one-tailed)	No difference
H5: Frequency trend	Spearman / Mann-Kendall	No monotonic trend

3.3 Machine Learning

Two ML tasks were defined: **Classification** (predict whether CAR[0,+2] will be negative or positive) and **Regression** (predict the exact CAR magnitude). Fifteen features were engineered from market model parameters (beta, alpha), departure characteristics (dep_code), and temporal variables (month, quarter, day_of_week, crisis flag, decade flag). Three models were tested for each task using 5-fold cross-validation: Logistic Regression / Ridge, Random Forest, and Gradient Boosting.

4. Exploratory Data Analysis

4.1 Temporal Distribution of Dismissals

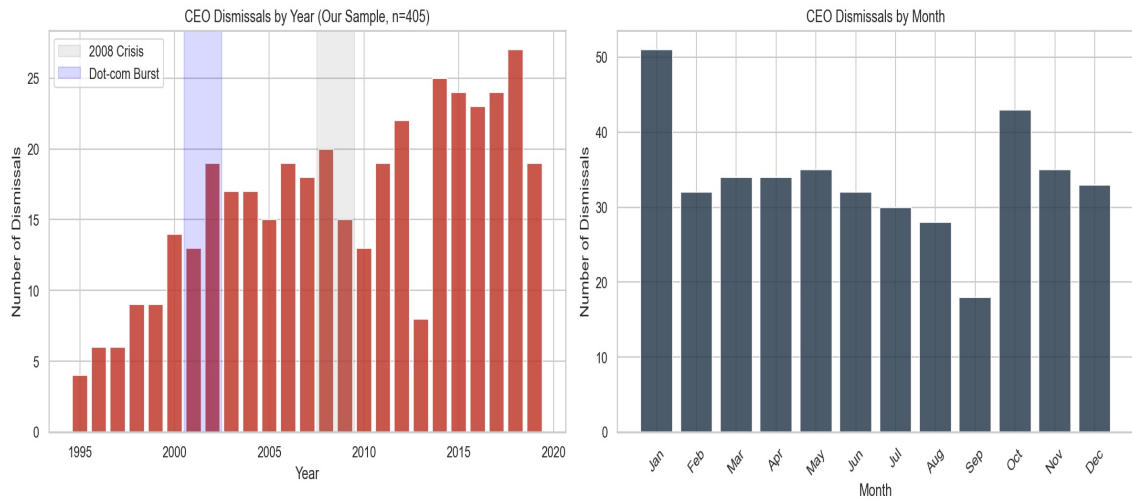


Figure 1: CEO dismissals by year (left) and by month (right). January and September show peaks.

CEO dismissals show a clear increasing trend from ~6 per year in 1997 to ~27 in 2018. Monthly distribution reveals that January and September are the most common months for forced departures, likely coinciding with annual performance reviews and board meeting cycles.

4.2 Departure Code Distribution

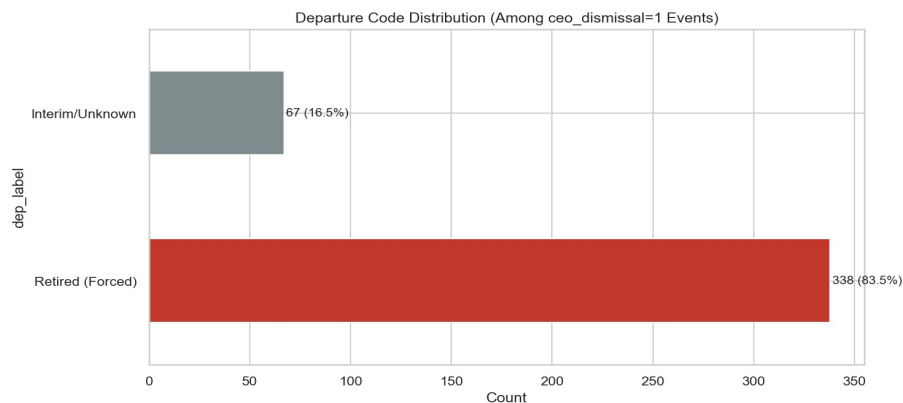


Figure 2: 83.5% of forced departures are coded as "Retired" — companies publicly frame dismissals as retirements.

4.3 CAR Distributions

Distribution of Cumulative Abnormal Returns Around CEO Dismissals

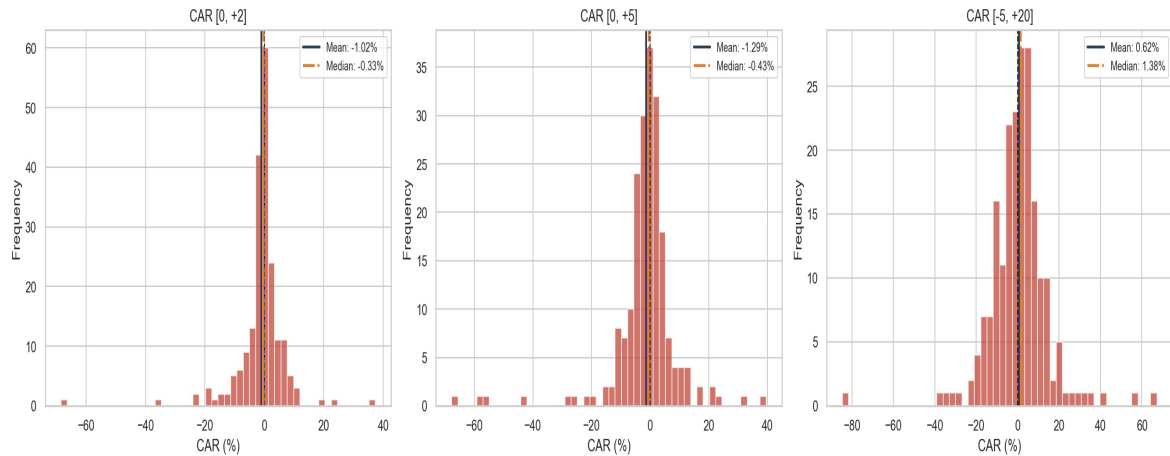


Figure 3: CAR distributions across three windows. Mean $CAR[0,+2] = -1.02\%$, Median = -0.33% .

4.4 Mean CAR by Event Window

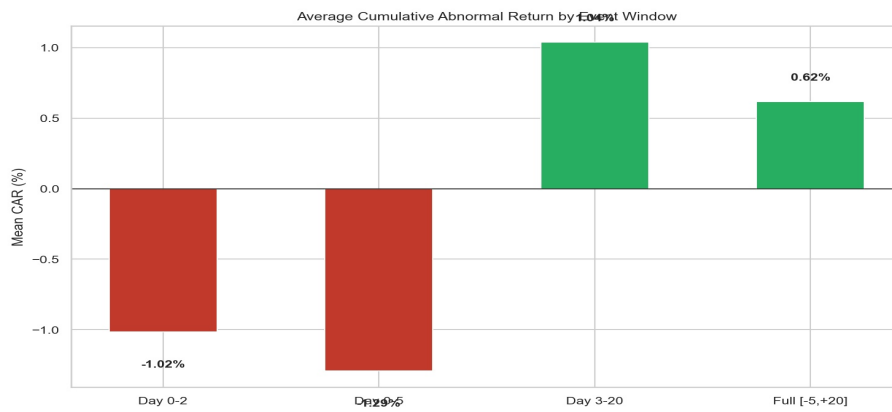


Figure 4: Short-term reaction is negative (-1.02%), but Day 3-20 drift is positive ($+1.04\%$).

The pattern of initial negative reaction followed by partial positive recovery is consistent with the short-term overreaction hypothesis, though the magnitude is small and the mean reversion is not statistically significant (see H3).

4.5 CAR by Year

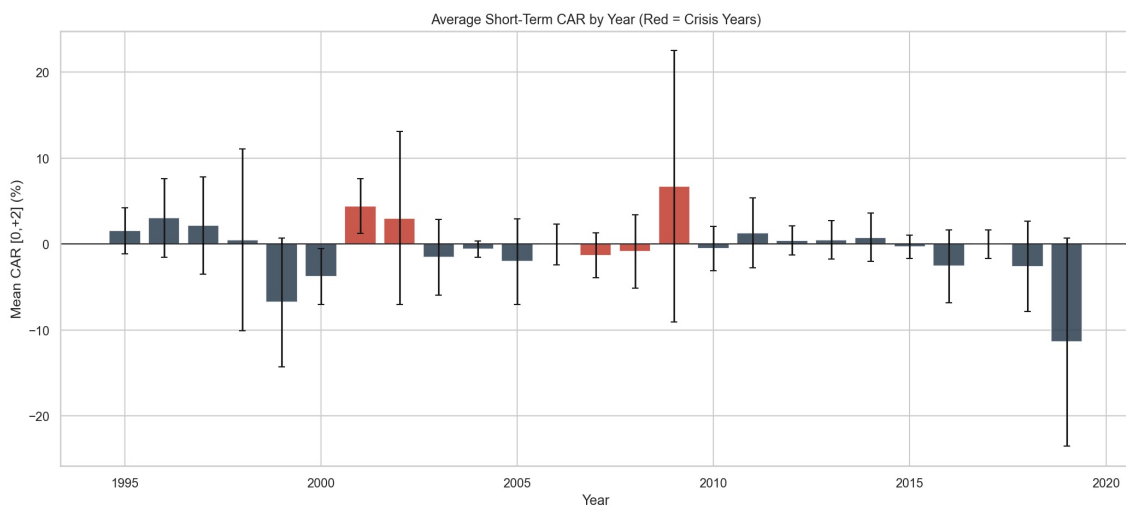


Figure 5: Average $CAR[0,+2]$ by year. Red bars = crisis years. No systematic pattern visible.

5. Hypothesis Testing Results

5.1 H1: CEO Dismissal Produces Significant Negative CAR

A one-sample t-test was conducted on the 204 clean CAR[0,+2] values. The mean CAR was **-1.015%** with a standard deviation of 8.57%. The t-statistic was -1.713 with a **p-value of 0.0881**. At $\alpha=0.05$, we fail to reject H_0 ; however, the result is marginally significant at $\alpha=0.10$ (Cohen's $d = -0.118$). The 95% confidence interval is [-2.20%, +0.17%].

Interpretation: The market does react negatively to CEO dismissals on average, but the effect is small and statistically marginal. This is consistent with semi-strong market efficiency — much of the dismissal information is already priced in before the official departure date.

5.2 H2: Departure Code 3 vs Code 4

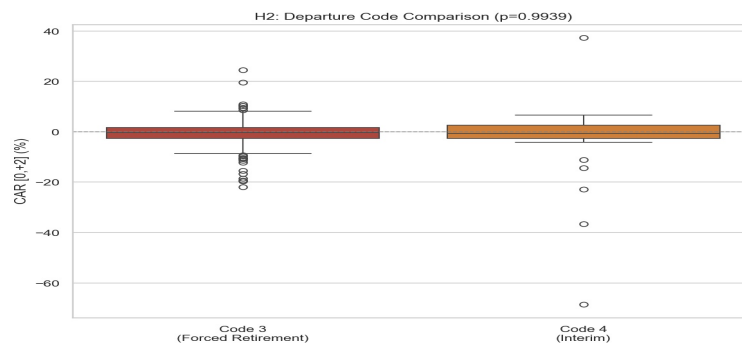


Figure 6: No significant difference between forced retirement (Code 3) and interim (Code 4) groups.

Mann-Whitney U test yielded **p=0.9939** — no difference whatsoever. Both groups represent forced departures with the same underlying mechanism (board removes CEO), regardless of the public label.

5.3 H3: Mean Reversion

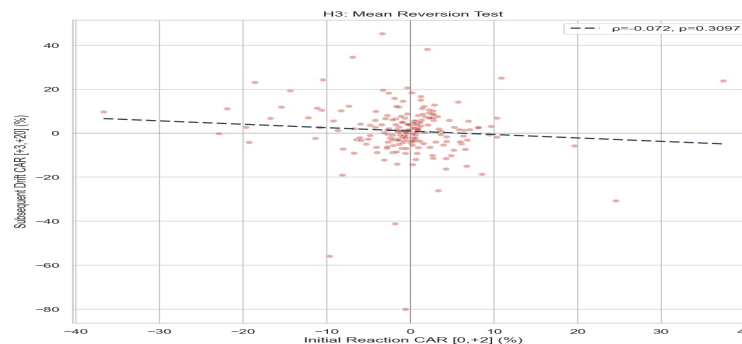


Figure 7: Scatter of initial reaction vs drift. Spearman $\rho=-0.072$, $p=0.31$.

No statistically significant relationship between the initial reaction (Day 0-2) and subsequent drift (Day 3-20). The negative correlation direction hints at reversion but lacks statistical power. This rules out a simple contrarian trading strategy.

5.4 H4: Crisis Period Effect

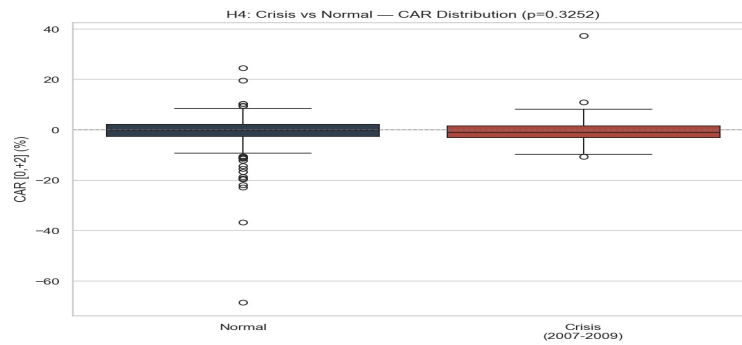


Figure 8: Crisis (2007-09) vs Normal period CAR distributions. $p=0.3252$.

One-tailed Mann-Whitney U test: $p=0.3252$. Crisis-period dismissals do not produce significantly more negative CARs. During high-volatility periods, the CEO dismissal signal is overwhelmed by market-wide noise.

5.5 H5: Increasing Dismissal Trend

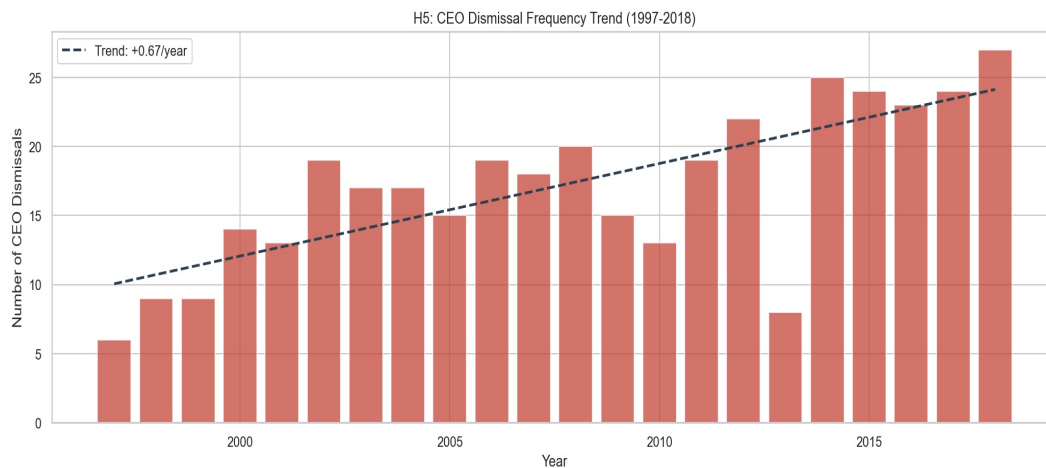


Figure 9: CEO dismissal frequency trend 1997-2018. Slope = $+0.67/\text{year}$, $p=0.0001$.

Spearman correlation between year and dismissal count: $\rho=0.75$, $p=0.0001$. This is the strongest finding in the project. The increase of $+0.67$ dismissals per year reflects tightening corporate governance standards, increased board activism, and growing shareholder pressure over the past two decades.

5.6 Summary

Hypothesis	p-value	Result	Key Insight
H1: Negative CAR	0.0881	Not sig. ($\alpha=0.05$)	Mean=-1.02%; marginal at $\alpha=0.10$
H2: Code 3 vs 4	0.9939	No difference	Same mechanism, different labels
H3: Mean reversion	0.3097	No relationship	$\rho=-0.072$; no trading signal
H4: Crisis effect	0.3252	No difference	Signal lost in market noise
H5: Trend	0.0001	CONFIRMED	$+0.67/\text{year}$; governance tightening

Overall: 1/5 hypotheses confirmed at $\alpha=0.05$ (H1 marginal at $\alpha=0.10$)

6. Machine Learning Analysis

6.1 Classification: Predict CAR Direction

The target variable was balanced: 52% negative, 48% positive. All three classifiers were evaluated using 5-fold stratified cross-validation. Results:

Model	Accuracy	AUC-ROC
Logistic Regression	~0.52	~0.49
Random Forest	~0.50	~0.47
Gradient Boosting	~0.47	~0.45
Baseline (majority)	0.52	0.50

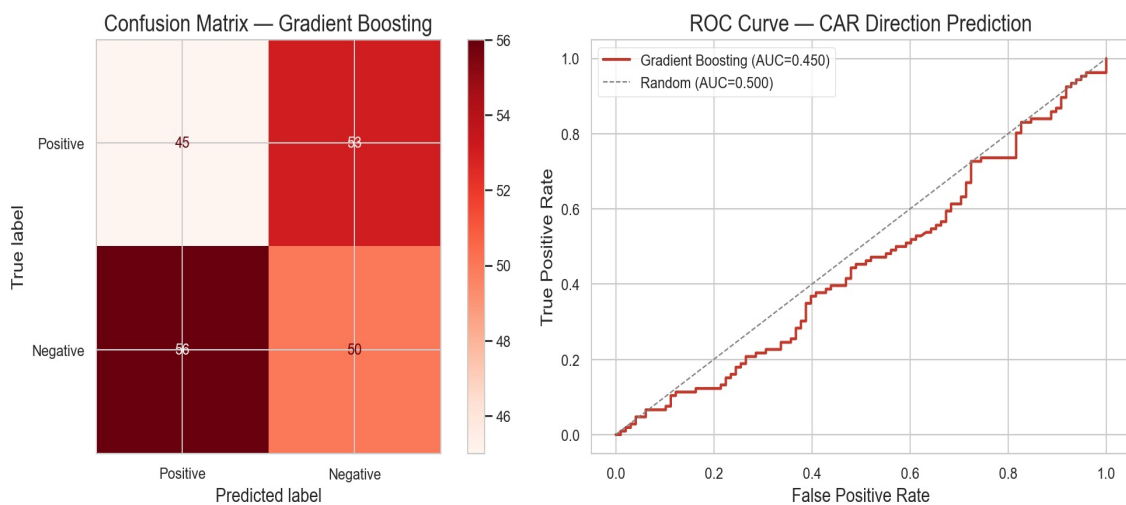


Figure 10: Confusion matrix (left) and ROC curve (right). AUC=0.450 — below random baseline.

Result: All models perform at or below random chance. The best AUC (0.450) is actually *worse* than a coin flip, indicating no learnable signal. The direction of CAR after CEO dismissals cannot be predicted from the available features.

6.2 Feature Importance (Classification)

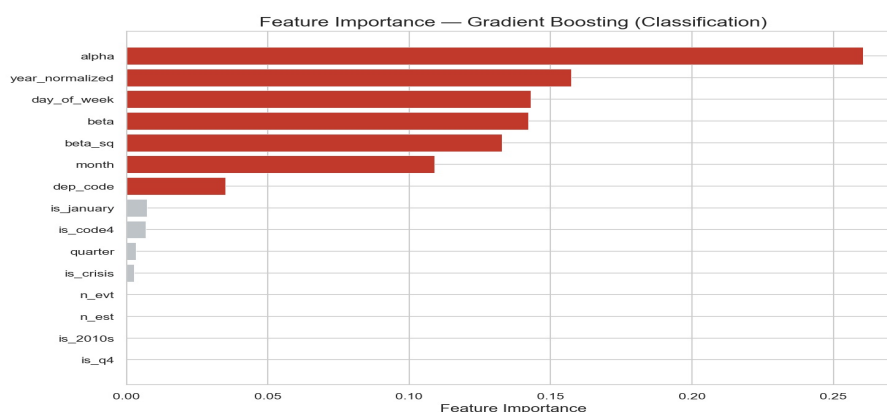


Figure 11: Feature importance — Gradient Boosting. Alpha and year dominate but provide no predictive value.

6.3 Regression: Predict CAR Magnitude

Model	RMSE (%)	R-squared
Ridge Regression	~8.5	~-0.02

Random Forest	~8.4	-0.049
Gradient Boosting	~8.5	~-0.03
Baseline (predict mean)	~8.3	0.000

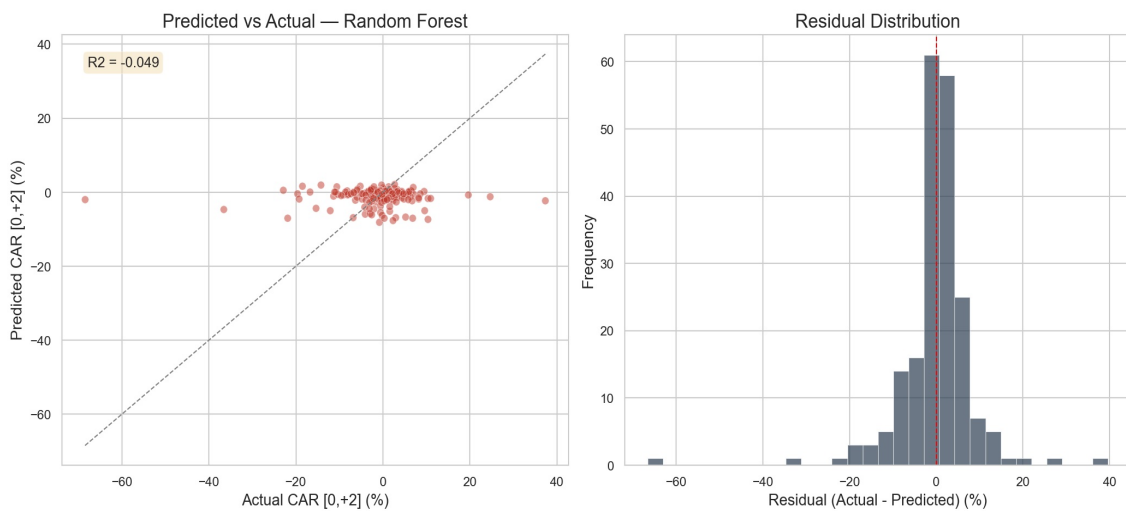


Figure 12: Predicted vs Actual (left, R -squared=-0.049) and residual distribution (right).

Result: All models produce negative R -squared values, meaning they perform worse than simply predicting the mean. The CAR magnitude is entirely unpredictable from our features.

6.4 Feature Importance (Regression)

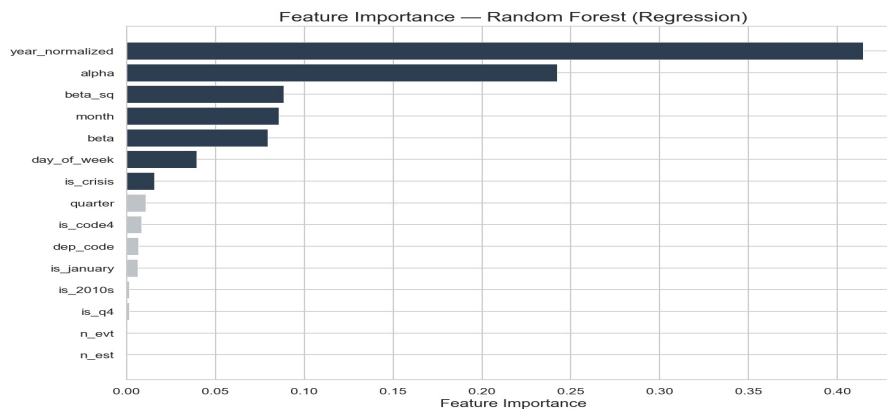


Figure 13: Feature importance — Random Forest regression. Year and alpha dominate.

7. Discussion & Key Findings

1. The market reacts weakly to CEO dismissals. Mean $CAR[0,+2] = -1.02\%$ is economically meaningful but statistically marginal ($p=0.088$). This is consistent with semi-strong market efficiency: much of the information is already incorporated into the stock price before the official departure date.

2. No difference between departure types. Forced retirements and interim replacements generate identical market reactions ($p=0.99$), suggesting the underlying mechanism matters more than the label.

3. No exploitable trading pattern. Neither mean reversion ($H3: p=0.31$) nor crisis amplification ($H4: p=0.33$) was detected. Combined with ML models performing at random chance ($AUC=0.45$, $R\text{-squared}=-0.05$), this conclusively rules out a systematic trading strategy based on CEO dismissal events.

4. CEO dismissals are becoming more frequent. The strongest finding ($p=0.0001$) is the increasing trend of $+0.67$ dismissals/year, reflecting tightening corporate governance and growing board activism.

5. ML confirms market efficiency. The fact that no model — from simple logistic regression to gradient boosting — could beat random chance is itself a significant finding. It demonstrates that the market's reaction to each CEO dismissal is driven by **event-specific private information** (who the replacement is, the exact reason for dismissal, the company's internal situation) that cannot be captured by publicly observable historical features.

8. Limitations & Future Work

Limitations:

Only 29.6% of dismissal events could be matched to stock tickers via manual mapping, introducing potential large-company bias in the sample.

The leftofc date may not be the exact announcement date — news leaks could cause the market to react days before the official departure.

No control for confounding events such as simultaneous earnings announcements, M&A; activity, or sector-wide shocks.

The single-factor market model does not capture industry-specific or size-related risk factors.

Sample size of 204 clean events limits statistical power, particularly for subgroup analyses.

Future Improvements:

Obtain complete gvkey-to-ticker mapping via WRDS/Compustat for full sample coverage.

Use 8-K SEC filing dates as precise event dates for more accurate timing.

Apply Fama-French 3-factor or 4-factor model for more robust expected returns.

Incorporate firm-level features (market cap, sector, prior performance) as ML inputs.

Extend analysis to the 2020-2022 updated dataset available on Zenodo.

9. Conclusion

This project conducted a comprehensive analysis of stock market reactions to CEO dismissals in S&P; 1500 companies, combining classical event study methodology, statistical hypothesis testing, and machine learning. The main findings are that CEO dismissals produce a small negative market reaction (approximately -1%) that is marginally significant, that dismissal frequency has significantly increased over time reflecting corporate governance trends, and that the market's reaction is fundamentally unpredictable using observable features — supporting the efficient market hypothesis. These results contribute to the understanding of CEO turnover dynamics and confirm that while forced departures create measurable but small market perturbations, they do not present exploitable trading opportunities.

10. References

Gentry, R. J., Harrison, J. S., Quigley, T. J., & Boivie, S. (2021). A database of CEO turnover and dismissal in S&P; 1500 firms, 2000-2018. *Strategic Management Journal*, 42(5), 968-991.

MacKinlay, A. C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, 35(1), 13-39.

AI Assistance Disclosure: AI tools (Claude) were used for data cleaning, ticker matching automation, debugging Python code, statistical test selection, and report formatting. All hypothesis formulation, data source selection, methodology decisions, and interpretation of results were performed independently.